Unit 07: Data Link Layer-medium access protocol

CONTENTS

Objectives

- understand bit-oriented protocol.
- know the frame format and types of HDLC and PPP frame.
- understand multiple access protocol.

Introduction

The frame is clearly viewed as a set of bits with no semantics or context in a bit based approach. Regardless of the frame contents, a bit oriented protocol can switch data frames. HDLC is a bit-oriented protocol.

HDLC

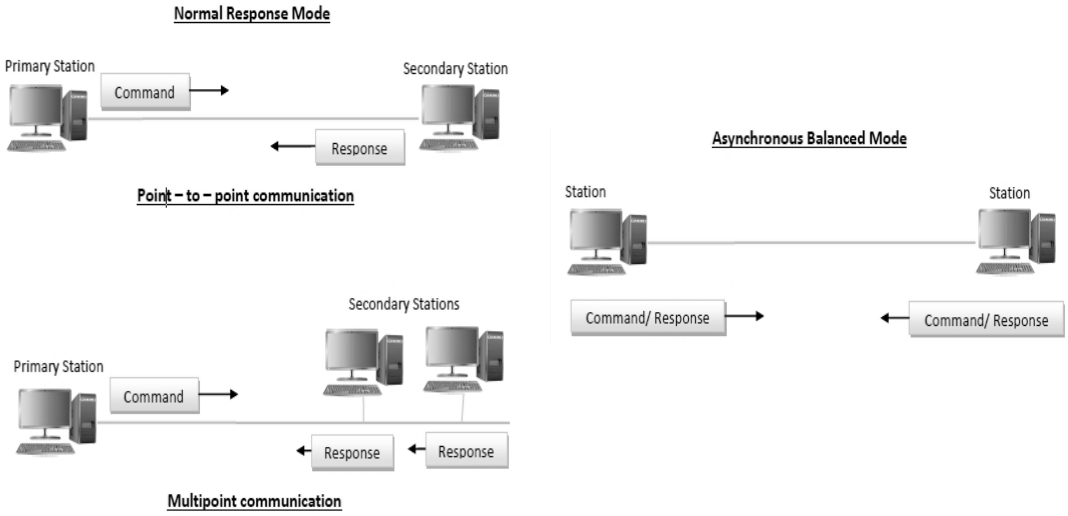
HDLC stands for high-level Data Link Control protocol, which ensures that data frames are delivered reliably over a network or communication link. HDLC also offers framing, data transparency, error detection and correction, and even flow management, among other things.

7.1 History of HDLC

Let's look at the history of HDLC. The synchronous Data Link Control protocol, developed by IBM, is an example of a bit based protocol. The SDLC is currently standardised as HDLC synchronous Data Link Control, who standardises as an HDLC. The SDLC was standardised by the ISO as the high level Data Link Control protocol, and SDLC is really important for the data link layer. It is not commonly used, but it is critical for many data link layer protocols. It's a bit-oriented protocol that can be used for point-to-point and multi-point communication.

Transfer mode of HDLC

As you can see in the Figure 1, HDLC has two transfer modes: normal response mode and asynchronous balanced mode. There are two kinds of stations in normal response mode. A primary station sends orders, and a secondary station responds to the commands sent. In point-to-point and multi-point correspondence, the normal response mode is used. Asynchronous Balanced mode is when the setup is balanced, which ensures that each station can send and receive commands. It is only used for one-to-one correspondence.



HDLC Frame format

The first is the beginning sequence, which is 8 bits long and shows you how many bits will be used to indicate the start and end of the frame as seen in . This is also known as the flag bits, and it consists of eight bits, meaning that this is the start of the series. After that, there's a 16-bit header. Your body sections, which are variable in volume, are the third one. We can't define the scale since the body is so big. This body one would receive whatever data it receives from the network layer. The CRC stands for error detection with cyclic redundancy scan, and it has a 16-bit duration. Then comes the ending sequence, which is eight bits long and indicates the end of the frame; the starting and ending sequences are both 06 times one and none. This sequence is also sent while the connection is idle such that the sender and receiver's clocks remain in sync. The starting and ending sequences are 0, 6 times one, then zero.

8	16		16	8
Beginning Sequence	Header	Body	CRC	Ending Sequence

Figure 2 HDLC Frame format

During some times when the connection is idle, this sequence is also transmitted. Why? So that the sender and receiver can keep their clocks synchronized. The next field is the header, which I've already showed you, followed by the address and control fields, which hold the address and control foot. And there's the body, which is a variable-size payload of whatever data we're getting from the satellite. The next section is the shell, which is a variable-size payload information that any data we get from the network layer will be inserted into this body one CRC cyclic redundancy search error detection mechanism. We've already seen that any data is collected from the network layer in the datalink layer is added to the header and the trailer trailer, which are the error detection bits and CRC. So, header refers to the header, and trailer header refers to the trailer header, which has a 16-bit CRC. A trailer is what we call it in this country. Error correction parts will be used in the teaser.

Types of HDLC Frame

There are two fields in the HDLC frame format header: one is known as the address field, and the other is known as the control field. So, what is the purpose of a control field? The purpose is to define the type of HDLC frame.

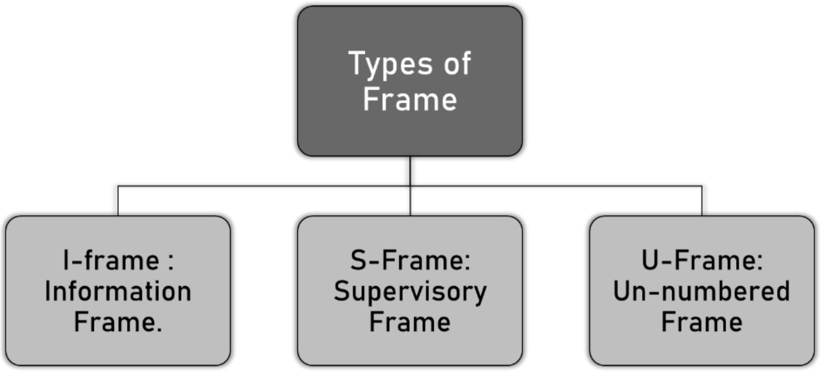


Figure 3 Types of Frames

It's as if you're trying to figure out what kind of HDLC frame you have. The control field determines the frame class, and the control field is, in turn, a part of the header. So there are three different kinds of HDLC frames: I-frame, S-frame, and U-frame as shown in Figure 4. The knowledge frame is also known as the I-frame. A supervisor frame is also known as a S frame. Unnumbered frames are also known as U-frames.

I-Frame	**1^{st} bit is 0**
S-Frame	1^{st} two bits is 10
U-Frame	1^{st} two bits is 11

Figure 4 Types of HDLC Frames

The control field determines the frame forms. As you can see in the table, we have an I-frame, an S-frame, and a U-frame. An I-frame is one in which the first bit in the control field is zero; an S-frame is one in which the first bit in the control field is one zero; and a U-frame is one in which the first two bits or one zero is in the control field. It's a S frame in a U frame, which stands for unnumbered frame. A unnumbered frame is one in which the first bit is one.If the first bit in the control field is zero, the frame is an Iframe, which means it carries information. If the first two bits in the control field are one, zero, the frame is a supervisor frame.So, the supervisor frame plays a part in error management and flow control systems, and the third one is the unnumbered frame. It's used for a variety of tasks, including connection management.

Byte Oriented Protocol

Consider the frame as a string of bytes or a character. That is why, as we discussed in the previous lecture, it is often regarded as a character-oriented approach.

In general, we have three protocols.

BISYNC -> PPP -> Binary Synchronous Communication Protocol

PPP -> Point-to-Point Protocol.

DDCMP -> Message Protocol for Digital Data Communication

7.2 Point-to-Point Protocol

It is a data link layer protocol that is available. The Point to Point Protocol (PPP) is a van protocol that is widely used over internet connections. When we say "internet connections," we're referring to two routers. The Point to Point Protocol (P2P) is a commonly used protocol. It's also known as the van protocol. Since there are two major applications for the Point to Point Protocol.the two most popular applications It's commonly used in internet networks that have a lot of traffic and a lot of speed. Obviously, the internet is one of the stuff that has a high load and a fast pace at the same time. It's often used to send data from two multipoint-to-point computers, or two computers that are directly connected.

Components of PPP

- Encapsulation Component
- Link Control Protocol (LCP)
- Authentication Protocols (AP)
- Network Control Protocols (NCPs)

Encapsulation Component

It encapsulates data grams in order for them to be transferred over this physical layer.

Link Control Protocol (LCP)

It is in charge of creating, configuring, checking, maintaining, and terminating connections, as well as negotiating ties. It also makes use of functionality across two end point connections.

Authentication Protocols (AP)

Password authentication protocol and challenge Handshake Authentication Protocol, also known as PAP protocol and CHAP protocol, are two authenticated protocols of Point to Point Protocol that authenticate endpoints for usage of services. The acronyms PAP and CHAP stand for Password Authentication Protocol and Challenge Handshake Authentication Protocol, respectively.

Network Control Protocols (NCPs)

The conditions and services for the networks are negotiated using this protocol.

Frame format of PPP

PPP frames are typically used to encapsulate packets of data or information that only contain configuration data or data. PPP is based on the same fundamental format as HDLC. PPP usually has one more field, the protocol field. After the control field and before the input or data field, this protocol field appears.

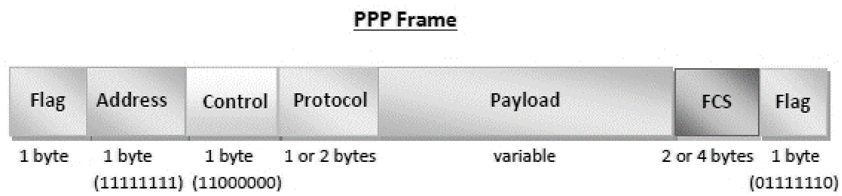


Figure 5 Frame format of PPP

Flag field –

PPP frames are identical to HDLC frames in that they both begin and end with the regular HDLC flag. It always has a 1-byte value, which is 01111110 in binary.

Address filed –

The address field essentially serves as a transmitted address. All 1's basically means that all of the stations are able to embrace frame in this case. It has a 1-byte value, or 11111111 binary value. Specific station addresses are not given or assigned by PPP, on the other hand.

Control filed –

In HDLC, this area primarily uses the U-frame (Unnumbered frame) format. The control field is needed for various purposes in HDLC, but in PPP, it is set to 1 bit, or the binary value 00000011. This 1 byte is used for a data link that does not need a connection.

Protocol field –

This area essentially defines the datagram's network protocol. It usually determines the type of packet in the data field, or what is being transported in the data field. This field is 1 or 2 bytes long and is used to identify the PDU (Protocol Data Unit) encapsulated by the PPP frame.

Data filed –

It normally includes the datagram from the upper layer. For standard PPP data frames, the network layer datagram is especially encapsulated in this region. The length of this field varies rather than being continuous.

FCS filed –

This area normally includes a checksum for the purpose of detecting errors. It can be 16 bits or 32 bits in length. It's even done on the address, access, protocol, and also information fields. Characters are applied to the frame for monitoring and error management.

Byte Stuffing in PPP frame

When the flag sequence appears in the packet, byte stuffing is used in the PPP payload field such that the recipient does not believe it to be the end of the frame. Each byte that comprises the same byte as the flag byte or the escape byte has the escape byte, 01111101, packed before it. Before transferring the message into the network layer, the receiver extracts the escape byte.

7.3 Multiple Access Protocols in Computer Network

The Data Link Layer is responsible for transmission of data between two nodes. Its main functions are-

- Data Link Control
- Multiple Access Control

Data Link control –

The data link control is responsible for reliable transmission of message over transmission channel by using techniques like framing, error control and flow control. For Data link control refer to – Stop and Wait ARQ

Multiple Access Control –

If the sender and receiver have a dedicated connection, the data link control layer is sufficient; but, if the sender and receiver do not have a dedicated link, several stations will reach the channel at the same time. To reduce collisions and prevent crosstalk, multiple access protocols are needed.When an instructor asks a question in a classroom full of students, and all of the students (or stations) begin responding at the same time (send data at the same time), a lot of confusion is generated (data duplication or data loss). It is then the teacher's task (multiple access protocols) to control the students and get them answer one at a time.For data sharing on non-dedicated networks, protocols are thus necessary. Multiple control protocols can be further broken down into the following subcategories:

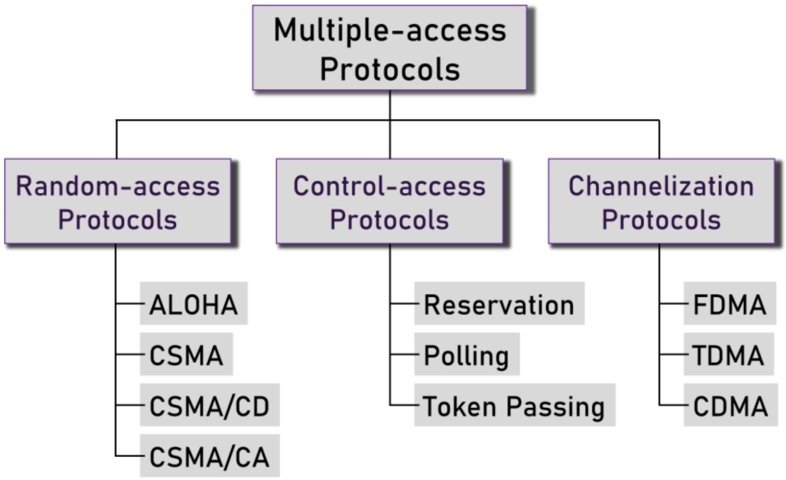


Figure 6 Multiple Access protocols

7.4 Random-access protocols

It manages station connection to the transmission line. It sends out a connection that necessitates the use of an access control system. As a result, random access protocols are used. We use a variety of methods. So, Aloha CSMA, CSMA CD, and CSMA CA, respectively.

(a) ALOHA

It was created for a wireless LAN, but it can also be used for a shared medium. Multiple stations will relay data at the same time in this scenario. This is why, in this situation, we are confronted with a crash and data that is jumbled. If you can see in the Figure 7, if two cars begin transmitting data at the same time or begin using the station at the same time, there is a risk of a crash.

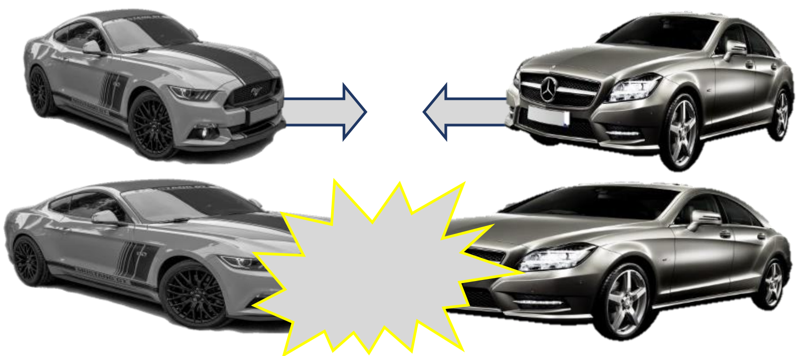


Figure 7 Collision

If more than one station begins using the channel at the same time, a collision can occur. There are two separate interpretations of aloha. One is pure Aloha. The second is slotted ALOHA as shown in Figure 8.

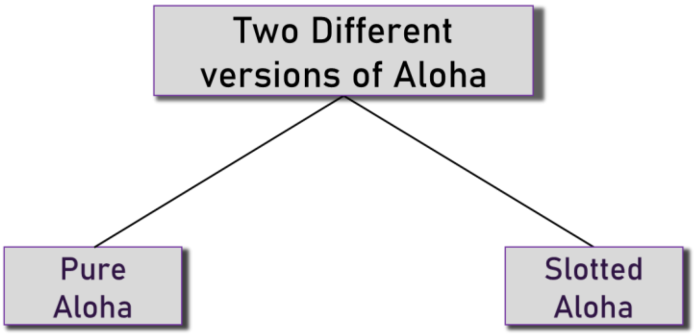


Figure 8 Different versions of ALOHA

Pure Aloha

When a station transmits data, it waits for a response. If the acknowledgment does not arrive within the allotted time, the station waits a random period of time (Tb) before re-sending the results. Since multiple stations take varying amounts of time to wait, the chances of another collision are reduced.

Vulnerable Time = 2* Frame transmission time

Throughput = G exp{-2*G}

Maximum throughput = 0.184 for G=0.5

Slotted Aloha

It's equivalent to pure aloha, except that we split time into slots and data can only be sent at the start of each slot. If a station runs out of time, it must wait for the next available slot. This lowers the chances of a crash.

Vulnerable Time = Frame transmission time

Throughput = G exp{-*G}

Maximum throughput = 0.368 for G=1

Difference between Pure Aloha and Slotted Aloha

The difference between pure aloha and slotted aloha is shown in Table 1.

Table 1 Pure Aloha vs Slotted Aloha

Pure Aloha	Slotted Aloha
In Pure Aloha, any station can transmit data at any time.	In Slotted Aloha, any station can transmit data only at beginning of any time slot.
In Pure Aloha, time is continous and is not globally syncronized.	In Slotted Aloha, time is discrete and is globally syncronized.
Vulnerable time for pure aloha = 2 x T_{fr}	Vulnerable time for Slotted aloha = T_{fr}
In Pure Aloha, Probability of successful transmission of data packet= G x e^{-2G}	In Slotted Aloha, Probability of successful transmission of data packet= G x e^{-G}
Pure aloha doesn't reduce the number of collisions to half.	Slotted aloha reduces the number of collisions to half and doubles the efficiency of pure aloha.
Maximum efficiency = 18.4% (Occurs at g = ½)	Maximum efficiency = 36.8% (Occurs at g = 1)

(b) CSMA

Since the station must first sense the medium (for idle or busy) before transmitting data, Carrier Sense Multiple Access means less collisions. It sends data if the channel is idle; otherwise, it waits for the channel to become idle. However, because of the propagation delay, there is also a risk of a collision in CSMA.

For example: Station A will first feel the medium before sending results. It will begin transmitting data if the channel is found to be idle. If station B asks to send data and senses the medium, it will also find it idle and send data by the time the first bit of data is sent (delayed due to propagation delay) from station A. As a consequence, data from stations A and B will collide.

CSMA access methods: -

1-persistent: The node detects the channel and sends the data if it is idle; otherwise, it continuously checks the medium for idleness and transmits unconditionally(with 1 probability) when the channel becomes idle.

Non-persistent: If the channel is idle, the node sends the data; if not, it tests the medium after a random period of time (not continuously) and transmits when it is found idle.

P-persistent: The node detects the medium and, if it is idle, sends data with a probability of p. If the data isn't sent ((1-p) chance), it waits a while and then scans the medium again; if it's already idle, it sends with p probability. This process is repeated before the frame is submitted. Wifi and packet radio networks both use it.

O-persistent: The superiority of nodes is determined ahead of time, and transmission takes place in that order. Node waits for its time slot to transmit data if the medium is idle.

(c) CSMA/CD

Carrier detects multiple entry points and detects collisions. If a collision is observed, stations may stop data transmission.

(d) CSMA/CA

Multiple access is detected by the carrier, and collisions are avoided. The sender receives recognition signals as part of the collision detection process. The data is successfully transmitted if there is only one signal (its own), so if there are two signals (its own and the one with which it collided), a collision has occurred.

7.5 Controlled access

In monitored access, the stations exchange data to determine which station has the authority to transmit. To stop message collisions on shared medium, it only requires one node to send at a time.

The three controlled-access methods are:

- Reservation
- Polling
- Token Passing

Reservation

Until sending info, a station must make a reservation using the reservation process. There are two types of periods on the timeline:

1. A fixed-length reservation interval

2. Variable frame data transfer time

Polling

The polling procedure is similar to a roll call in class. A handler, like the coach, sends a message to each node in turn.

One serves as the main station (controller), while the others serve as secondary stations. Both data must be exchanged via the controller. The address of the node being chosen for access is included in the message received by the controller.

While all nodes receive the message, only the one to which it is sent responds and sends data, if any. If there is no evidence, a “poll reject” (NAK) message is normally returned as shown in Figure 10 and Figure 9.The polling messages have a high overhead, and the controller's reliability is highly dependent.

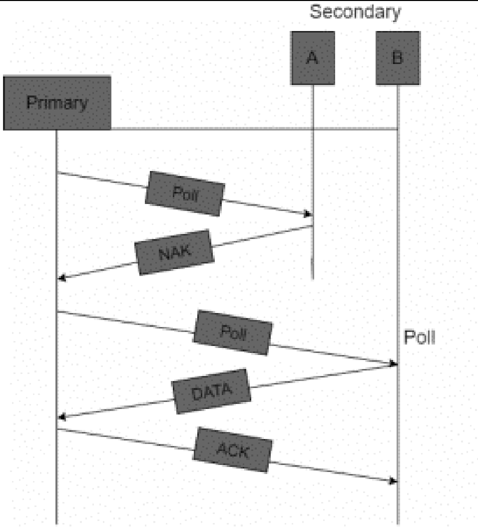


Figure 10 Polling (a)

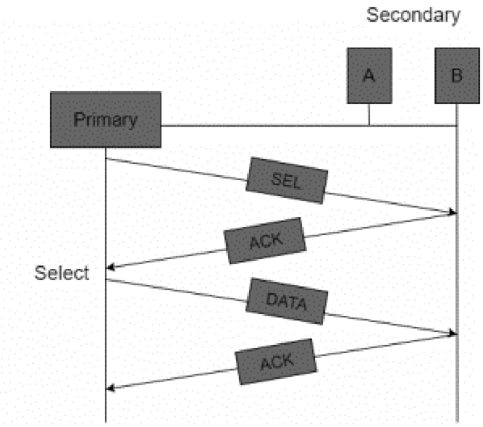


Figure 9 Polling (b)

Token passing

The stations in a token passing scheme are theoretically bound to one another in the form of a loop, and station access is controlled by tokens.

A token is a small message or a special bit pattern that circulates from one station to the next in a predetermined order.

Tokens are exchanged from one station to the next in the ring in the case of Token ring, while in the case of Token bus, each station uses the bus to transfer tokens to the next station in a predetermined order.

In all instances, the token denotes the ability to submit. When a station receives the token and has a frame queued for transmission, it will transfer the frame before passing the token to the next station. If there is no queued loop, it merely transfers the token as seen in Figure 11.

Following the transmission of a frame, each station must wait for all N stations (including itself) to send the token to their neighbors, as well as the other N – 1 station to send a frame if they have one.

There are issues such as token duplication or loss, insertion of a new station, and replacement of a station that must be addressed in order for this scheme to operate correctly and reliably.

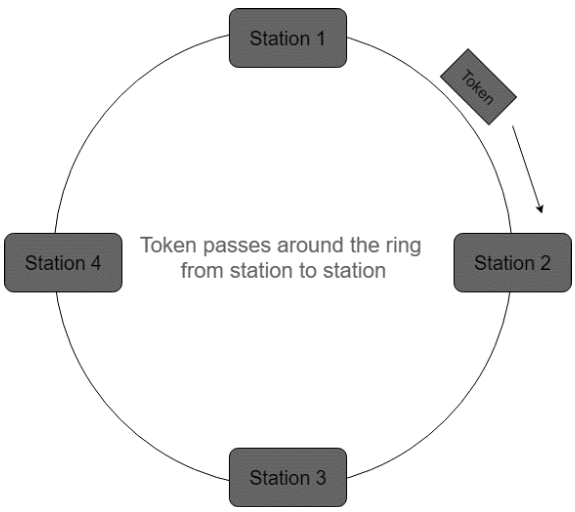


Figure 11 Token Passing

7.6 Channelization Protocols

The usable bandwidth of the link is shared in time, frequency, and code among multiple stations to allow them to access the channel at the same time.

FDMA (Frequency Division Multiple Access)

The usable spectrum is split into equivalent bands, allowing each station to have its own band. To stop crosstalk and disruption, guard bands have been added to ensure that no two bands overlap.

TDMA (Time Division Multiple Access)

The bandwidth is used by a number of different stations. To prevent collisions, time is split into slots, and stations are assigned to relay data during these slots. However, there is a synchronisation overhead since each station must know the time slot. Add synchronisation bits to each slot solves this issue. Another problem with TDMA is propagation delay, which can be overcome by using guard bands.

CDMA (Code Division Multiple Access)

Both signals are carried on a single channel at the same time. There is no such thing as bandwidth or time division. If there are several people in a room speaking at the same time, for example, flawless data reception is also possible if only two people talk the same language. Similarly, data from several stations can be broadcast in various code languages at the same time.

Summary

- The data link layer is divided into two sub layers. The upper sub layer is in charge of data link management, while the lower sub layer is in charge of addressing mutual media connectivity.
- The Finite State Machine model is a technique for verifying the protocol's correctness. The data link protocols PPP and HDLC are commonly used.
- Bluetooth is a proprietary open wireless technology protocol for transmitting data over short distances (using short wavelength radio communications in the ISM band from 2400 to 2480 MHz) between fixed and mobile devices, allowing for the development of highly secure personal area networks (PANs).
- In CSMA/CA, the sender collects the acknowledgement to identify potential collisions, and if there is just one acknowledgement present (its own), the data-frame has been transmitted successfully.
- Until submitting info, a station must make a reservation using the reservation access system. Intervals are used to divide time. A reservation frame precedes the data frames sent in that interval in each interval.

Keywords

Point-to-Point (PPP): It's a data link layer protocol that links two connecting link-level peers at either end of the link over a point-to-point link.

Aloha: Multiple access (MA) to the shared medium is possible with ALOHA. This arrangement has the potential for collisions. When one station sends data, another can try to send data at the same time. The data from the two stations clash, resulting in a jumbled mess.

CSMA: Carrier sense multiple access with collision detection (CSMA/CD) adds collision detection to the CSMA algorithm. After sending a picture, a station tracks the medium to see if the transmission was accurate. If that's the case, the station is over. If there is a collision, though, the frame is sent again.

Controlled access: In controlled access, the stations confer to determine who has the authority to deliver. It is not possible for a station to submit until it has received permission from other stations.

Channelization: Channelization is a multiple-access system in which a link's usable bandwidth is exchanged between separate stations in terms of time, distance, or code.

FDMA: The usable spectrum is split into frequency bands in frequency-division multiple access (FDMA). A band is assigned to each station for data transmission. To put it another way, each band is assigned to a particular station and remains with that station at all times.

CDMA: To gain multiple access in code-division multiple access (CDMA), the stations use separate codes. CDMA is based on coding theory and employs number sequences known as chips.

Self Assessment

1. .... describes the techniques to access a shared communication channel and reliable transmission of data frame in computer communication environment.
2. .... does not include any connection setup or release and does not deal with frame recovery due to channel noise.
3. .... refers to a reliable transfer of bit streams to the network layer for which the data link layer breaks the bit stream into frames.
4. .... controls mismatch between the source and destination hosts data sending and receiving speed and therefore dropping of packets at the receiver end.
5. We have categorized access methods into _________ groups
   a. Two
   b. Three
   c. Four
   d. Five
6. In _______, the stations share the bandwidth of the channel in time.
   a. FDMA
   b. CDMA
   c. TDMA
   d. none of the above
7. In the _______ method, the stations in a network are organized in a logical ring.
   a. Polling
   b. token passing
   c. reservation
   d. none of the above
8. _________ augments the CSMA algorithm to detect collision.
   a. CSMA/CD
   b. CSMA/CA
   c. either (a) or (b)
   d. both (a) and (b)
9. In the _________ method, after the station finds the line idle, it sends its frame immediately. If the line is not idle, it continuously senses the line until it finds it idle.
   a. p-persistent
   b. non persistent
   c. 1-persistent
   d. none of the above
10. In the _____ method, time is divided into intervals. In each interval, a reservation frame precedes the data frames sent in that interval.

a. token passing
b. Reservation
c. Polling
d. none of the above

11. In the _______ method, each station has a predecessor and a successor.

a. token passing

b. polling

c. reservation

d. none of the above

12. The vulnerable time for CSMA is the ________propagation time.

    a. three times

    b. two times

    c. the same as

    d. none of the above

Answers for Self Assessment

1.	Data link layer	2.	Unacknowledged connectionless sevice	3.	framing	4.	Rate of data transmission	5.	A
6.	C	7.	B	8.	A	9.	C	10.	B
11.	A	12.	C						

Review Questions

1. What is the data link protocol?
2. List three categories of multiple access protocols.
3. How can a collision be avoided in CSMA/CD network?
4. Compare and contrast CSMA/CD and token passing access methods.
5. Is Slotted Aloha always better than Aloha? Explain your answer with justification.
6. How does PPP transmit data grams over serial point-to-point links?
7. How does PPP establish link for authenticated transfer of file?
8. What are different data link protocols available? Why does PPP have become popular?
9. How does the data link layer accomplish the transmission of data from the source network layer to the destination network layer?
10. Compare and contrast a random-access protocol with a channelizing protocol.
11. Do we need a multiple access protocol when we use the local loop of the telephone company to access the Internet? Why?
12. Define channelization and list three protocols in this category

Further Readings

Andrew S. Tanenbaum, *Computer Networks*, Prentice Hall.

Behrouz A. Forouzan and Sophia Chung Fegan, *Data Communications and Networking*, McGraw-Hill Companies.

J. D. Spragins, *Telecommunications Protocols and Design*, Addison-Wesley.